SY B. TECH CSE (AI/TECH) • SEMESTER III

AI and ML Essentials

Unit 1 : Introduction to Artificial Intelligence (AI)

Course Faculty : Ms.Priyanka A.Hajare

Course Quick Facts

Parameter	Course Specifications	Academic Classification
Course Code	ICS23PCL201	Department Core Curriculum
Course Category	PCC (Professional Core Course)	Computer Science & Engineering Specialization
Teaching Scheme	**L-3** Hrs / Week	3 Lectures per week (3 Credits equivalent)
Theory Exam Duration	**2 Hours**	Standardized End-Semester Theory Evaluation

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Prerequisites & Stats Baseline

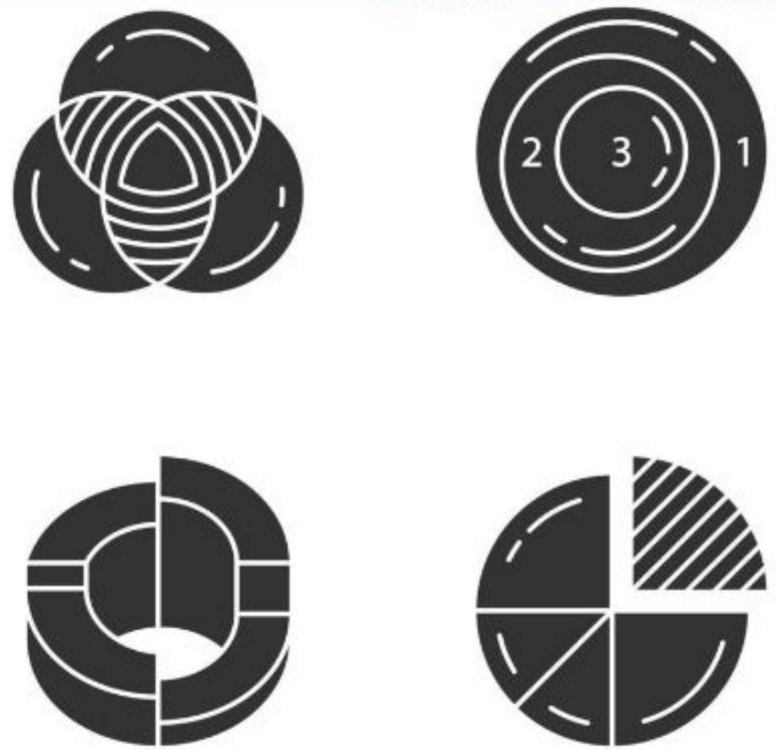
Why Statistics is the Foundation

Machine Learning and AI models require solid familiarity with statistical patterns and probability distributions to make predictions under uncertainty.

Key mathematical components needed:

- Probability & Bayes' Rule
- Descriptive & Inferential Statistics

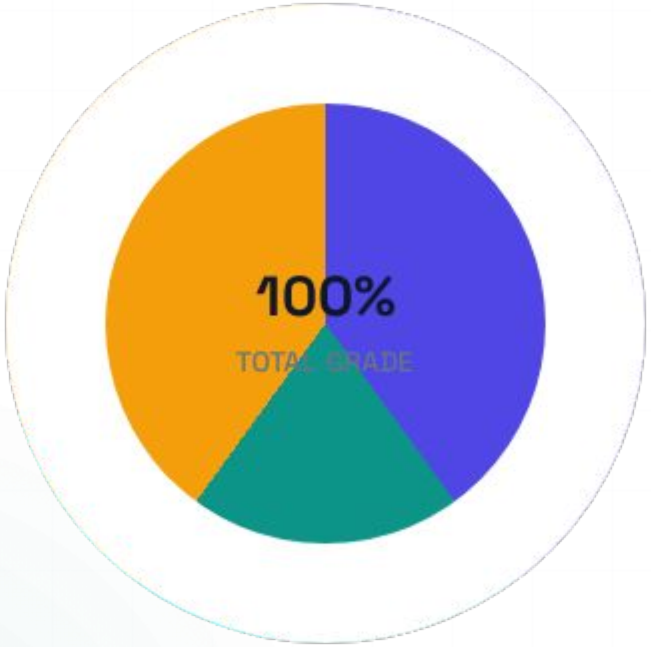
Bayes' Rule: $PA \mid B = \frac{PB \mid A \cdot PA}{PB}$



Course Objectives

1. To introduce fundamental concepts and historical evolution of Artificial Intelligence.
2. To explain logical reasoning, knowledge representation, and inference mechanisms in AL.
3. To develop understanding of probabilistic reasoning and Bayesian methods.
4. To familiarize students with evolutionary algorithms and optimization techniques.
5. To present essential machine learning models and learning paradigms.

Course Evaluation Scheme



- Continuous Assessment (CA) — 40%
- Mid-Sem Exam (MSE) — 20%
- End-Sem Exam (ESE) — 40%

Continuous Assessment (CA) combined with Mid-Semester Exam (MSE) comprises 60% of the course weight, emphasizing consistent, steady engagement over late-stage cramming.

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Course Outcomes

What You Will Master

CO1: Explain fundamental principles of classic AI.

CO2: Apply logical & statistical reasoning to real AI problems.

CO3: Analyze evolutionary algorithms for complex optimization.

CO4: Distinguish key ML models & structural learning paradigms.

CO5: Demystify the structural execution of Neural Network paths.

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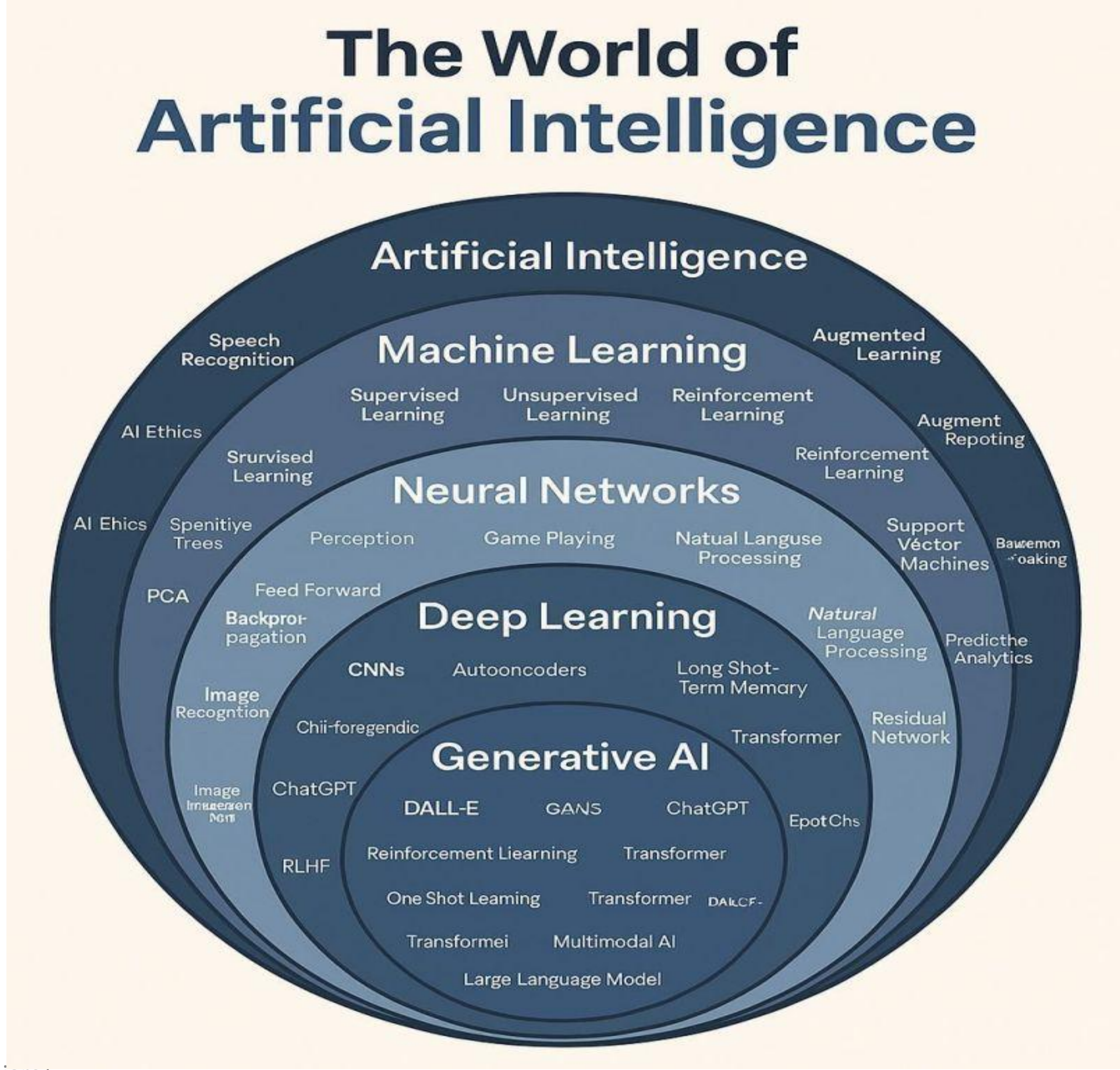
Unit 1 Introduction to Artificial Intelligence (AI)

Content

- Introduction to Artificial Intelligence
- Early AI Concepts
- Key Milestones in AI Development
- Influential AI Contributors
- Logical Approach to AI
- Propositional Logic and First-Order Logic
- Knowledge Representation
- Rule-Based Systems
- Knowledge-Based Systems
- Inference Engines
- Case-Based Reasoning
- Applications of Artificial Intelligence

- What is AI
- Artificial Intelligence (AI) is a technology that enables machines and computers to perform tasks that typically require human intelligence. It allows systems to learn from data, recognize patterns, and make decisions to solve complex problems



Defining AI: The 4 Quadrants

Russell & Norvig categorize historical definitions of Artificial Intelligence into four distinct dimensions:

Thinking Humanly

COGNITIVE

The exciting new effort to make computers think... machines with minds, in the full and literal sense. Explored via cognitive modeling.

Thinking Rationally

RATIONALLY

The study of mental faculties through the use of computational models. Focused strictly on the "laws of thought" (Logic).

Acting Humanly

BEHAVIORAL

The art of creating machines that perform functions that require intelligence when performed by people. Assessed by Turing's Test.

Acting Rationally

AGENTIC

The study of the design of rational agents. A rational agent is one that acts so as to achieve the best expected outcome.

Early AI Concepts & Key Milestones

John McCarthy is widely recognized as the **Father of Artificial Intelligence**. An American computer scientist and mathematician, McCarthy officially **coined the term "Artificial Intelligence"** in 1955 in his proposal for the historic Dartmouth Workshop, which took place in 1956. This conference brought together top minds and established AI as an independent academic discipline.

McCarthy believed that every aspect of learning or intelligence could, in principle, be so precisely described that a machine could be made to simulate it. Beyond naming the field, he made massive technical contributions, including inventing **LISP** (the programming language that dominated AI research for decades) and developing **time-sharing systems**, which allowed multiple people to use a single computer simultaneously.

While McCarthy holds the official title, the creation of AI was a collaborative effort, and a few other pioneers are frequently celebrated alongside him:

- **Alan Turing:** Often called the **Father of Modern Computer Science**. He provided the essential philosophical blueprint for AI with his 1950 paper computing machinery and intelligence and the Turing Test.
- **Marvin Minsky:** Co-founded the MIT AI laboratory with McCarthy. He was a pioneer of cognitive science and a leading authority on early neural networks and symbolic AI.
- **Herbert Simon & Allen Newell:** Created the **Logic Theorist** (1956), widely considered the first running AI program, which successfully proved complex mathematical theorems.

Early AI Concepts & Key Milestones

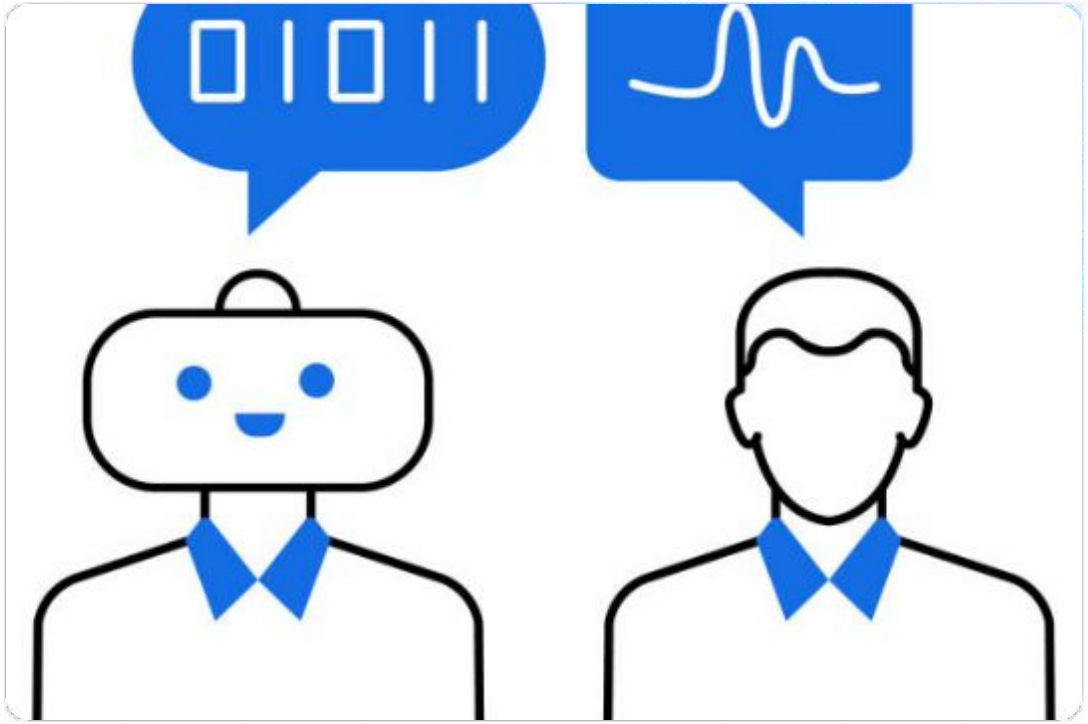
- **1950:** Alan Turing proposes the Turing Test for machine intelligence.
- **1956:** John McCarthy coins the term "Artificial Intelligence".
- **1956:** Logic Theorist program successfully solves complex mathematical theorems.

Turing's Operational Test

The Turing Test (1950)

Designed to provide a satisfactory operational definition of intelligence, sidestepping metaphysical debates on the nature of "thinking."

- **Natural Language Processing:** To communicate successfully in a human language.
- **Knowledge Representation:** To store what it knows or hears.
- **Automated Reasoning:** To use stored info to draw conclusions.
- **Machine Learning:** To adapt to new circumstances and find patterns.

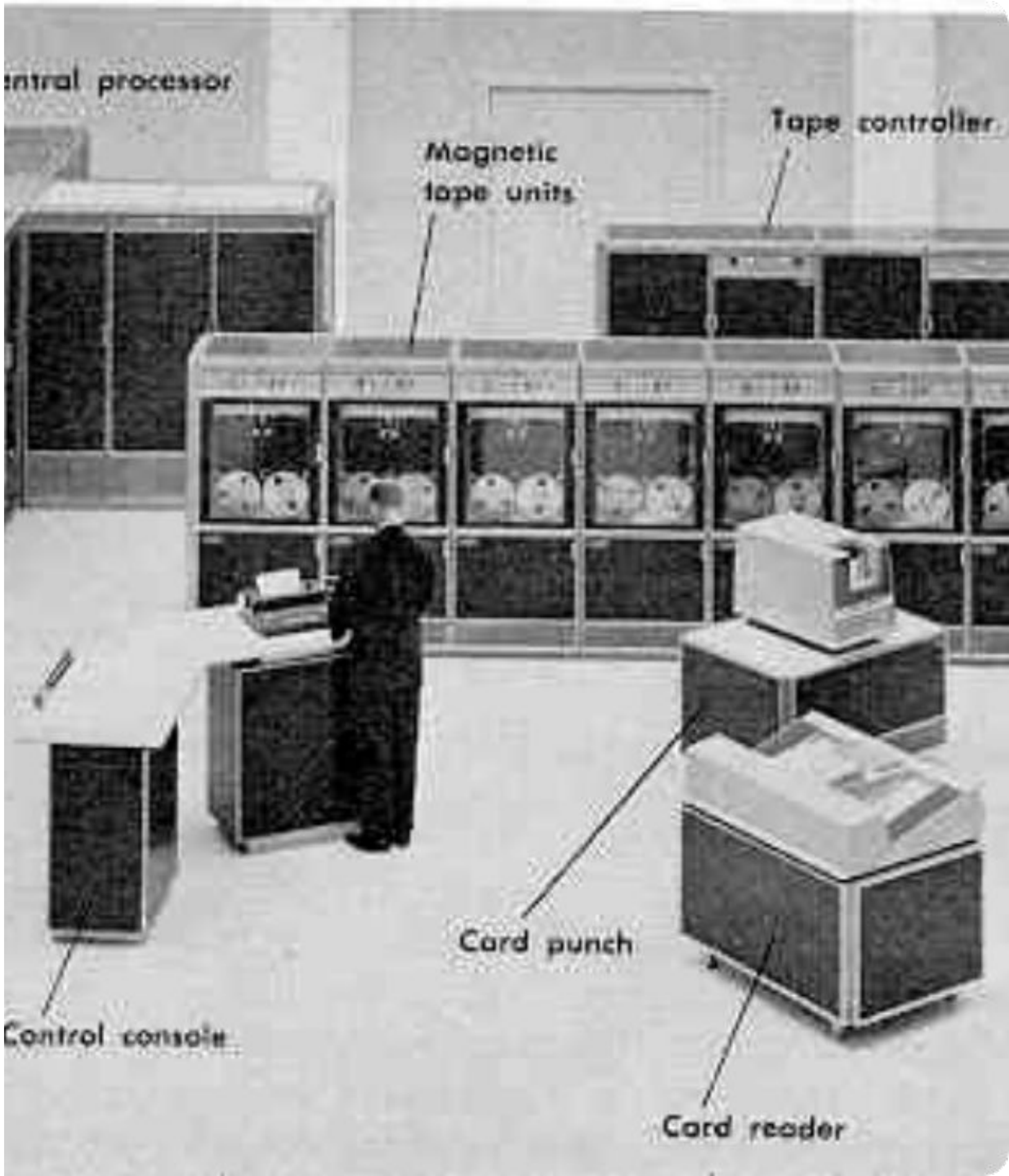


Gestation of Early AI

Cybernetics to Turing's Vision

The earliest formal work recognized as AI was done by Warren McCulloch and Walter Pitts (1943). They proposed a model of artificial neurons based on propositional logic, proving that any computable function could be implemented by a network of connected neurons.

Shortly after, in 1950, Alan Turing published his prophetic paper *"Computing Machinery and Intelligence"* which laid out the entire philosophical and operational landscape of AI today.



Timeline of Key Milestones

1943–1955
AI Gestation
McCulloch-Pitts neural networks, Turing's Turing Test proposal.

1956
Birth of the Field
McCarthy organises the Dartmouth Workshop, coining "AI".

1960s–1980s
First AI Winter
Overpromised goals led to funding loss, expert systems rise.

1990s–Present
Statistical AI
Introduction of probability, deep neural nets, big data.

Pioneers of AI Development

Alan Turing

Introduced the universal computing machine, the Turing Test, machine learning principles, and genetic algorithms.

John McCarthy

Organised Dartmouth, coined "Artificial Intelligence", designed the LISP programming language, and championed logical AI.

Marvin Minsky

Co-founded the MIT AI Lab, advanced cognitive frames, and analyzed artificial neural network limitations in "Perceptrons".

Simon & Newell

Built Logic Theorist (first AI program) and General Problem Solver, pioneering symbolic computer programming modeling.

The Logical Approach to AI

"Laws of Thought" Paradigm

This approach traces its roots back to Aristotle's codification of right thinking (syllogisms). It focuses on deriving correct, unassailable inferences.

Example: "Socrates is a man; all men are mortal; therefore Socrates is mortal." If premises are true, logic guarantees the conclusion is true.

Knowledge-Based Agent

Logic-based AI systems represent world knowledge explicitly through logic formulas. Sentences are parsed by an Inference Engine to plan actions.

This declarative paradigm decouples the representation of raw knowledge from the procedural algorithm driving the agent.

Fundamental AI Terminology with Examples

Agent

An autonomous entity that perceives its environment through sensors and acts upon it using actuators.

- *Example*: An Automated Self-Driving Car. It senses the road via LIDAR cameras and steers using its mechanical acceleration system.

Percept

The complete historical sequence of all sensory inputs an agent has received up to any given point.

- *Example*: A Smart Thermostat logging temperature readings over the last 24 hours: [72°F, 71°F, 75°F, 68°F].

Knowledge Base (KB)

A specialized, structured database containing sentences written in a formal representation language that stores facts, beliefs, and rules about the world.

- *Example*: A Medical Diagnosis System's KB containing rules like:
- Symptom(Flu, Fever) and \(\forall x (HasFever(x) \rightarrow IsSick(x))\).

Fundamental AI Terminology with Examples

Inference

The algorithmic process of calculating and deriving brand new logical sentences from existing facts within the Knowledge Base.

- *Example*: If a chatbot's KB contains IsRainy(Today) and IsRainy(Today) \rightarrow TakeUmbrella(Today), its inference engine derives the new instruction: TakeUmbrella(Today).

Semantics

The meaning of sentences; it defines the truth value of a sentence with respect to a specific layout of the world (model).

- *Example*: The sentence IsSunny(Today) has a semantic truth value of *True* if the actual physical sun is out, and *False* if it is storming outside.

Model

A formal mathematical world configuration that assigns specific truth values to every symbol in a language.

- *Example*: If your logic system uses two variables, {P: BatteryDead, Q: CarStarts}, one specific model is the world configuration where P = True and Q = False.

Fundamental AI Terminology with Examples

Entailment (double turnstile ⊩)

A relationship where one sentence follows logically from another.

If (A ⊩ B), then in every model where sentence (A) is true, sentence B *must* also be true.

- *Example*: The sentence [John is a CSE Student \wedge All CSE Students take Coding classes] entails the conclusion [John takes Coding classes].

Soundness

A property of an inference algorithm ensuring it will *only* derive sentences that are genuinely entailed by the Knowledge Base (it never generates false conclusions).

- *Example*: A sound mathematical calculator algorithm will never output `2 + 2 = 5`; it only produces logically valid, unassailable answers.

Completeness

A property of an inference algorithm ensuring it can successfully find and derive *every single sentence* that is logically entailed by the Knowledge Base.

- *Example*: A complete crime investigation AI will successfully find and name *all* possible suspects hidden within the police database without missing a single matching lead.

The Core Concept

In AI, an agent needs a way to think, plan, and handle situations it has never explicitly seen before. The Logical Approach treats the agent's mind as a structured repository of facts. Instead of hardcoding every action, we give the agent a set of baseline truths and rules. The agent then runs mathematical calculations (inference) over those rules to determine what to do next.

How a Knowledge-Based Agent Operates

A logic-based AI agent interacts with its world using a simple three-step execution loop:

1. **TELL:** The agent passes its current sensory percepts into the Knowledge Base to log new environmental facts.
2. **ASK**: The agent queries the Knowledge Base to figure out which action it should perform next based on its rules.
3. **TELL**: The agent commits to the chosen action and logs that decision into the Knowledge Base to keep track of its history.

Propositional Logic in AI

Propositional Logic is the simplest foundational logic system. It deals entirely with propositions, which are simple declarative statements that can only be either True or False. It treats each basic statement as a complete, indivisible unit (a "black box").

Syntax and Basic Inference

Propositional Logic deals with declarative facts (propositions) that a either true or false. Complex rules are structured using Boolean connectives.

- **Syntax:** Propositional variables represented by symbols (e.g., P, Q,
- **Connectives:** Negation, Conjunction, Disjunction, Implication, Biconditional.

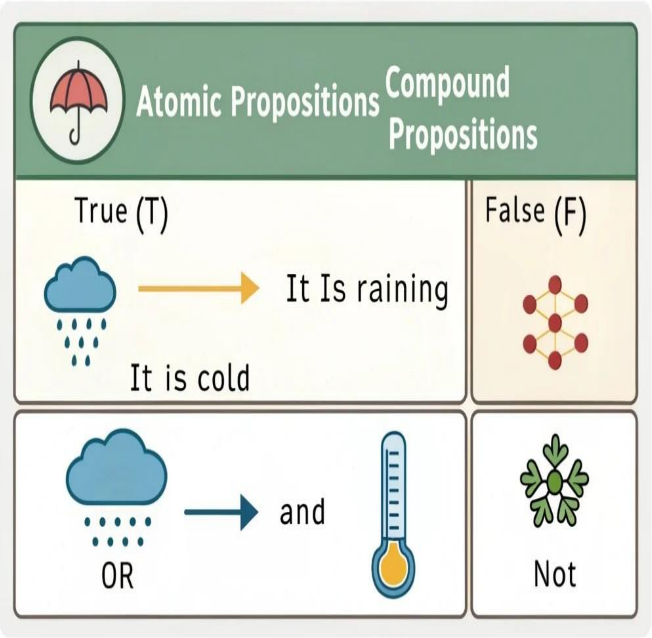
$$P \land Q \Rightarrow R$$

Connective Symbols	Words	Terms	Example
$\land$	AND	Conjunction	$A \land B$
$\lor$	OR	Disjunction	$A \lor B$
$\rightarrow$	Implies	Implication	$A \rightarrow B$
$\Leftrightarrow$	If and only If	Biconditional	$A \Leftrightarrow B$
$\neg$	Not	Negation	$\neg A$

Syntax (Building Blocks)

We use uppercase letters to act as Propositional Symbols (variables) like \(P, Q, R\). We join these variables together into complex sentences using five core Logical Connectives:

- Negation : The "NOT" operator. Flips a truth value.
- Conjunction : The "AND" operator. The whole sentence is true only if both sides are true.
- Disjunction : The "OR" operator. The sentence is true if at least one side is true.
- Implication : The "IF-THEN" operator. Asserts that if the first part is true, the second part must follow.
- Biconditional : The "IF-AND-ONLY-IF" operator. True only when both variables match in truth value.



First Order Logic (FOL)

First Order Logic (FOL) is an advanced logical system used in artificial intelligence that goes beyond propositional logic.

FOL represents knowledge as a collection of objects, attributes, and relations using the syntax: attribute/function(object(s)).

“first-order (or relational) representations” means to represent knowledge in such a way that it refers only to object properties and relations, and therefore generalizes across object identities.

FOL is a language—we define the syntax, the semantics, and give examples.

The limitation of propositional logic

- Propositional logic has nice properties:
  - Propositional logic is *declarative*: pieces of syntax correspond to facts
  - Propositional logic allows partial/disjunctive/negated information (unlike most data structures and databases)
  - Propositional logic is *compositional*: meaning of $B_{1,1} \wedge P_{1,2}$ is derived from meaning of $B_{1,1}$ and of $P_{1,2}$
  - Meaning in propositional logic is *context-independent* (unlike natural language, where meaning depends on context)

- Limitation:
  - Propositional logic has very limited expressive power, unlike natural language. E.g., we cannot express "pits cause breezes in adjacent squares" except by writing one sentence for each square

First-order logic

- Whereas propositional logic assumes that a world contains *facts*, first-order logic (like natural language) assumes the world contains
  - *Objects*: people, houses, numbers, theories, Ronald McDonald, colors, baseball games, wars, centuries . . .
  - *Relations*: red, round, bogus, prime, multistoried . . ., brother of, bigger than, inside, part of, has color, occurred after, owns, comes between, . . .
  - *Functions*: father of, best friend, third inning of, one more than, end of . . .

Comparing PL and FOL

Feature Specification	Propositional Logic (PL)	First-Order Logic (FOL)
Ontological Assumption	The world consists of absolute facts.	The world consists of objects, relations, functions.
Expressive Power	Lo LOW uires separate sentences for every fact.	Hi HIGH assert properties of groups via quantifiers.
Inference Complexity	Decidable (truth tables, SAT solvers).	Semi-decidable (First-order theorem proving).
Core Mathematical Concept	Truth-functional variables.	Quantifiers ($\forall$ $\exists$ $\forall$, $\exists$), Predicates, and Terms.

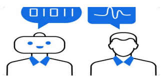
Case Study: Deep Blue

State Space Search & Heuristics

In 1997, IBM's chess computer, Deep Blue, defeated world champion Garry Kasparov. Deep Blue did not rely on modern deep learning. Instead, it was an expert symbolic logic system utilizing state-space adversarial search.

It evaluated up to 200 million chess board positions per second, combining recursive minimax alpha-beta pruning with handcrafted heuristic evaluation functions built by human grandmasters.

Image Sources



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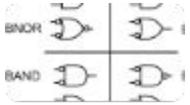

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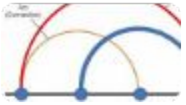

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LECTURE SERIES: AI & ML ESSENTIALS

Knowledge Systems & Expert Engineering

Exploring structural Knowledge Representation, Rule-Based logic, and the mechanics of human-like reasoning in Artificial Intelligence.

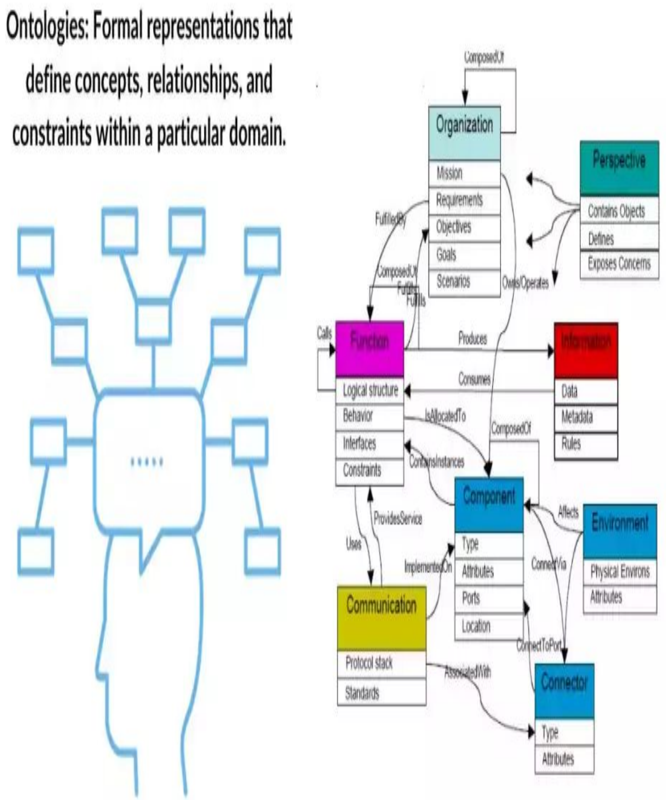
Introduction to Knowledge Representation

What is it?

- Knowledge Representation and Reasoning (KRR) are fundamental concepts in artificial intelligence (AI) that focus on how intelligent systems can effectively organise, store, and utilise knowledge. These two interrelated components enable machines to reason, make informed decisions, and simulate human-like cognitive processes.
- Human brains store facts, experiences, and rules to make daily decisions.
- Knowledge Representation (KR) is how we give this same information to a computer.
- It translates real-world knowledge into a format that a machine can understand and use.

The Core Goal:

- It is not just about storing data in a database.
- It is about structuring data so an AI can **reason** and **learn** from it.



IMG REF : https://spotintelligence.com/2024/01/16/knowledge-representation-and-reasoning-ai/

Understanding Knowledge Representation

The Essence of Knowledge Representation (2-Column Layout)

- **Column 1: Translating Complexity**
  - Maps complex real-world concepts into machine-readable structures.
  - Systematizes messy data by capturing entities, unique attributes, and deep structural relationships dynamically.

Column 2: Systematizing Information

- Creates structured pathways for AI logic frameworks.
- Allows systems to successfully navigate, manipulate, and utilize knowledge to drive automatic reasoning, problem-solving, and decision-making.

Core Frameworks of Representation

- **Symbolic Logic:** Uses precise symbols, logic gates, and math notation. Excellent for rigid, rules-based formal structures.
- **Semantic Networks:** Visual graphs linking object nodes via relationship edges. Highly intuitive for capturing complex, fluid associations.
- **Frames & Scripts:** Organizes data into predefined property slots. Ideal for modeling recurring real-world scenarios and context.
- **Ontologies:** Formal domain semantics mapping strict data constraints. Fosters data interoperability across separate AI systems.

4 Common Approaches in Knowledge Representation and Reasoning

Logical Representation :

This approach uses formal mathematical languages to encode facts and rules about the world. The most common forms are **Propositional Logic** (basic true/false statements) and **First-Order Predicate Logic** (which includes objects, properties, and relations).

- **How it Represents:** It uses strict symbols, constants, variables, and logical operators (AND, OR, NOT, IF-THEN).
- **How it Reasons:** The system uses inference rules like *Modus Ponens* to deduce new facts from existing rules.

Example:

- *Fact 1:* `Human(Socrates)` (Socrates is a human)
- *Rule:* `∀x (Human(x) → Mortal(x))` (For all x, if x is human, x is mortal)
- *Reasoned Conclusion:* `Mortal(Socrates)` (Therefore, Socrates is mortal)

4 Common Approaches in Knowledge Representation and Reasoning

Semantic Networks

A semantic network is a graphical approach that represents knowledge through a web of interconnected concepts. It was originally designed to model how human long-term memory stores words and ideas.

- **How it Represents:** It uses a network graph where **Nodes** represent objects or concepts, and **Edges (Arrows)** represent the explicit relationships between them.
- **How it Reasons:** The system traverses the graph arrows to inherit properties. If a specific node connects to a broader category, it automatically inherits the traits of that category.

Example:

- `[Canary]` $\xrightarrow{\text{is-a}}$ `[Bird]` $\xrightarrow{\text{has-part}}$ `[Wings]`
- *Reasoned Conclusion:* The system reasons that a Canary has wings, even though "wings" was never directly attached to the Canary node.

4 Common Approaches in Knowledge Representation and Reasoning

Frames

Proposed by AI pioneer Marvin Minsky, frames organize knowledge into structured, reusable packets or templates that describe typical real-world objects or situations.

- **How it Represents:** A frame is an object-oriented collection of **Slots** (attributes) and **Values** (the data inside those attributes). Slots can also have "default values" that are assumed true unless specified otherwise.
- **How it Reasons:** When the AI encounters a new object, it activates the matching frame template. It infers missing information by automatically falling back on the frame's default values.

Example:

- `Frame: Hotel Room`
    - *Slot: Bed → Value: King size (Default)*
    - *Slot: Bathroom → Value: Attached (Default)*

4 Common Approaches in Knowledge Representation and Reasoning

Production Rules

This is the most common approach used in classic expert systems and rule-based systems. It breaks knowledge down into distinct, actionable pieces of logic.

- **How it Represents:** It stores knowledge as a massive collection of declarative **IF-THEN** statements (Condition → Action).
- **How it Reasons:** The system uses an inference engine to match current real-world data against the "IF" conditions. When a match is found, the "THEN" action is triggered (or "fired"). This can happen via **Forward Chaining** (moving from data to a conclusion) or **Backward Chaining** (working backward from a goal to find supporting data).
- **Example:**
    - *Rule 1:* IF patient_temperature > 38°C THEN patient_has_fever = True
    - *Rule 2:* IF patient_has_fever = True AND patient_has_rash = True THEN diagnosis = "Measles"

KNOWLEDGE REPRESENTATION (KR) : Common Approaches in Knowledge Representation and Reasoning

Semantic Networks

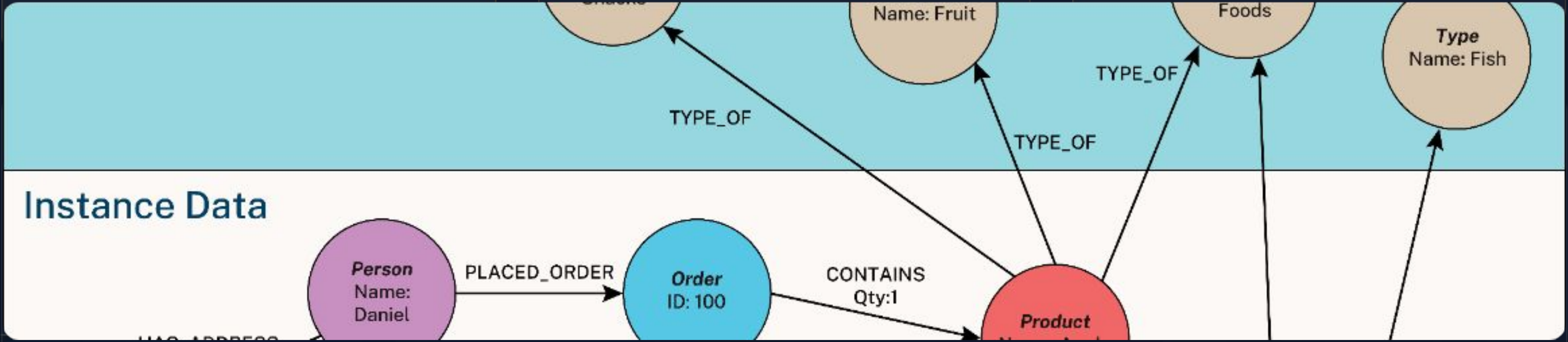
Representing knowledge as a graph of nodes (concepts) and labeled edges (relations). Enables **inheritance** of properties.

Frames & Scripts

Template-based structures with **slots** and **fillers** to represent stereotypical situations or objects.

Logic Systems

Formalizing facts using Propositional or First-Order Logic to ensure mathematically precise reasoning and deduction.



RULE-BASED SYSTEMS (RBS)

Production System Logic

Knowledge is represented as a set of **IF-THEN** rules (Production Rules). These mimic human expertise in specific domains.

```
IF ( Condition A ) THEN ( Action B )
```

Key Components

- **Working Memory:** Contains current facts.
- **Rule Base:** Long-term knowledge stored as rules.
- **Conflict Set:** Rules whose conditions are currently met.

Core Architecture of a Rule-Based System

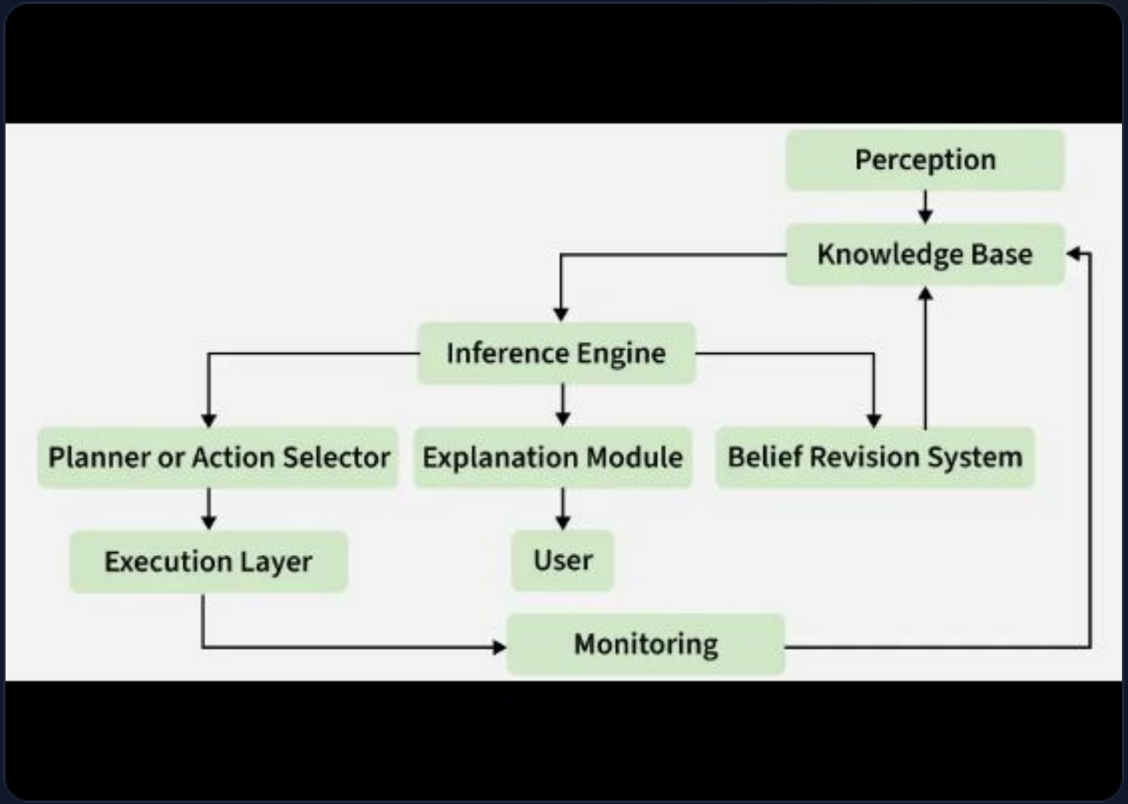
A typical system is divided into three main components:

Component	What it is	Example / Analogy
Knowledge Base	The **long-term memory** containing facts and expert "IF-THEN" rules.	*Rule:* "IF patient has a fever AND a sore throat, THEN consider strep throat."
Working Memory	The **short-term memory** storing the facts of the *current* situation.	*Data:* "Patient X has a fever. Patient X has a sore throat."
Inference Engine	The **brain** that matches the working memory data with the knowledge base rules to draw conclusions.	It matches the symptoms of Patient X to the rule and concludes: "Patient X might have strep throat."

KNOWLEDGE-BASED SYSTEMS (KBS)

The central concept of modern AI architecture: separating **What to know** from **How to use it**.

- **Knowledge Base (KB):** A domain-specific repository of facts and rules.
- **Inference Engine:** The brain of the system that applies rules to facts to derive conclusions.
- **User Interface:** Facilitates communication with the end-user.



Knowledge-Based System (KBS)

A **Knowledge-Based System (KBS)** is an AI program that uses a stored repository of human expertise to solve complex problems. Depending on how the knowledge is stored, organized, and processed, Knowledge-Based Systems are classified into several distinct types.

Types Knowledge-Based System (KBS)

1. Expert Systems (The Classic KBS)

This is the most common and traditional type of KBS. It mimics the decision-making ability of a human expert in a highly specific domain.

- How it works: It relies on a massive database of human-written IF-THEN rules and an inference engine to draw conclusions.
- Example: MYCIN, an early AI system used to identify blood infections and recommend antibiotics.

2. Case-Based Reasoning (CBR) Systems

Instead of relying on rigid rules, CBR systems solve new problems by looking at past examples or historical "cases." It operates on the human philosophy that "similar problems have similar solutions."

- How it works: When a new problem occurs, the system retrieves a similar past case from its memory, adapts that old solution to fit the new problem, and saves the new experience for future use.
- Example: Customer support help desks or legal research tools that look up past court cases to help lawyers build a new defense.

Types Knowledge-Based System (KBS)

3. Intelligent Tutoring Systems (ITS)

These are educational systems designed to provide personalized instruction and feedback to students without requiring a human teacher.

- **How it works:** The system contains a knowledge base of the subject matter, a model of the student's current learning flaws, and a set of teaching strategies to adapt the lessons dynamically.

Example: Language learning platforms like Duolingo or advanced AI math tutors like Carnegie Learning.

4. Hypertext Manipulation Systems

These systems manage large volumes of interconnected, textual information. They help humans find relationships across vast domains of written data.

How it works: It links concepts together using a web-like structure (similar to semantic networks), allowing users to easily navigate from one piece of knowledge to another related piece.

Example: Corporate internal wikis, online medical encyclopedias, or documentation search engines.

Types Knowledge-Based System (KBS)

5. Fuzzy Knowledge-Based Systems

Traditional systems only understand true/false or 1/0 logic. Fuzzy systems are built to handle the "gray areas" of human language and real-world uncertainty.

- **How it works:** It uses **Fuzzy Logic** to process imprecise words like "warm," "slightly," or "high" instead of exact numbers.
- **Example:** Smart washing machines that automatically adjust water levels based on how "dirty" the clothes look, or automated braking systems in cars.

6. Ontology-Based Systems

An ontology is a formal framework that defines the exact concepts, categories, and relationships within a specific industry.

- **How it works:** It links completely different databases together by forcing them to use a shared, standardized vocabulary. This allows different AI systems to share knowledge without confusion.
- **Example:** The **Semantic Web** or biomedical platforms that standardize thousands of protein and gene names across global research labs.

INFERENCE: FORWARD CHAINING

Data-Driven Reasoning

Starts with known facts and applies rules to extract all possible conclusions. It is **bottom-up** reasoning.

- Efficient for synthesis and monitoring tasks.
- May derive conclusions that are not relevant to a specific goal.
- Example: Real-time sensor monitoring in a factory.

Mechanism Flow:

Fact A & B → Rule 1 Fires → Infer Fact C

Fact C & D → Rule 2 Fires → Infer Fact E

Fact E → **Goal Reached**

INFERENCE: BACKWARD CHAINING

Goal-Seeking Flow:

Hypothesis H? ← Check Rule X

Condition G needed? ← Check Rule Y

Condition F needed? ← Found in Data!

Hypothesis Validated

Goal-Driven Reasoning

Starts with a goal (hypothesis) and works backwards to see if available data supports it. It is **top-down** reasoning.

- Efficient for diagnosis and classification.
- Focuses only on relevant data, ignoring the rest.
- Example: Medical diagnostic systems (MYCIN).

COMPARISON OF CHAINING METHODS

Feature	Forward Chaining	Backward Chaining
Direction	Data-driven (Bottom-up)	Goal-driven (Top-down)
Starting Point	Initial facts in Working Memory	Desired goal/hypothesis
Best Application	Planning, design, monitoring	Diagnosis, troubleshooting
Search Method	Breadth-first search (typically)	Depth-first search (typically)

CASE-BASED REASONING (CBR)

"Solving new problems by adapting solutions used to solve previous similar problems."

Retrieve

Find similar past cases from memory.

Reuse

Map the old solution to the new context.

Revise

Test and fix the solution as needed.

Retain

Store the new case for future use.

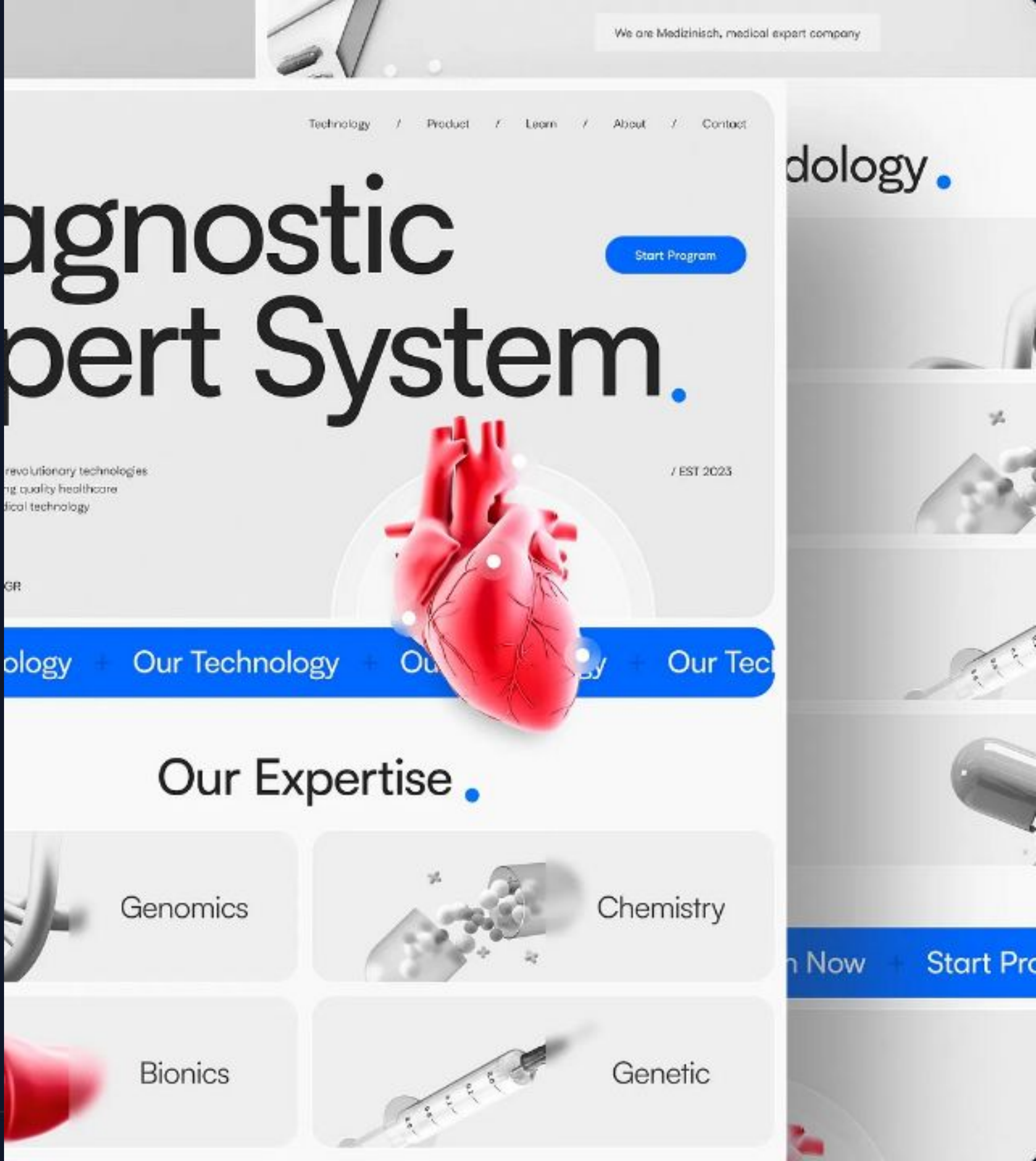
CASE STUDY: MYCIN

Diagnostic Expert System

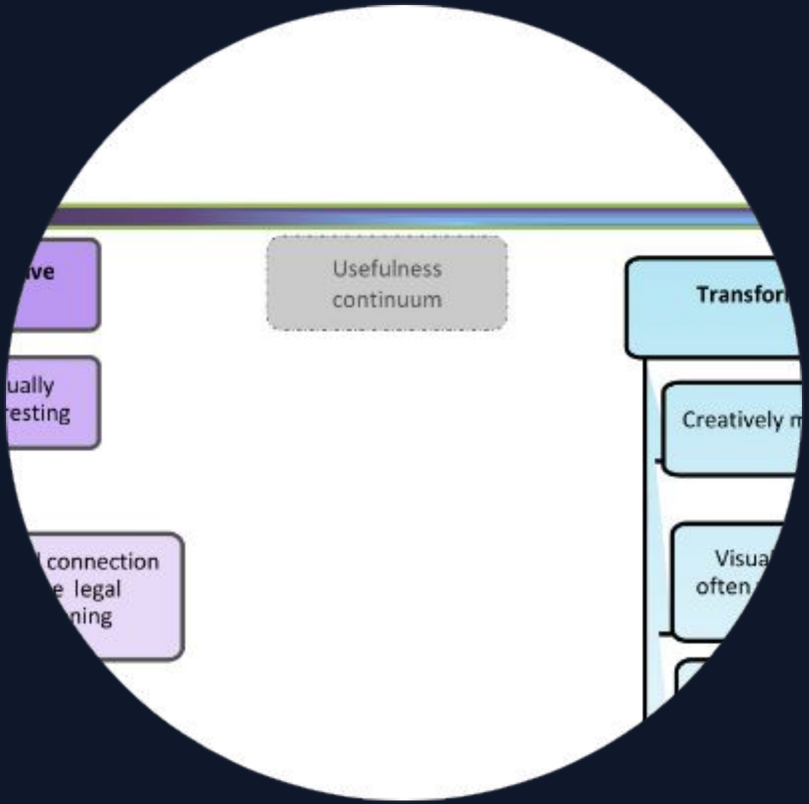
Developed at Stanford in the 1970s, MYCIN used roughly **600 rules** to identify blood infections and recommend antibiotics.

It utilized **Backward Chaining** with **Certainty Factors** to handle uncertainty, outperforming many non-specialist physicians in test scenarios.

- Knowledge Base: Infectious disease expertise.
- Inference: Goal-driven deduction.



CASE STUDY: LEGAL REASONING



Precedent-Based Decisioning

In legal systems, CBR mimics the judicial process of **Stare Decisis** (legal precedent).

Modern AI tools for lawyers "Retrieve" previous judgments on similar facts to suggest "Reuse" of successful arguments or "Revise" them based on current legislative changes.

Impact: Reduces research time from days to minutes, ensuring consistency in judicial interpretation.

BROAD APPLICATIONS OF AI

Healthcare

Computer-aided diagnosis, personalized treatment plans, and drug discovery.

Finance

Fraud detection, algorithmic trading, and credit risk assessment.

Robotics

Warehouse automation, autonomous vehicles, and surgical robots.

IMAGE SOURCES

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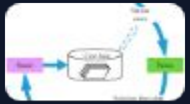

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